

Vincesid Omipon

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EDUCATION

University of British Columbia

Bachelor of Applied Sciences in Computer Engineering

Vancouver, BC

May 2025 – May 2029

Capilano University

Associate of Science

North Vancouver, BC

Sep. 2023 – May 2025

TECHNICAL SKILLS

Software: Java, Python, C/C++, SystemVerilog, HTML/CSS/JS, RISC-V & AVR Asm, Latex, Markdown, MS Office

Developer Tools: Git, VS Code, IntelliJ, Overleaf, Quartus, Questa, ModelSim, Linux, WSL, Ubuntu, AutoCAD, KiCad, LTSpice, MATLAB

Hardware: Oscilloscopes, Digital Multimeter, FPGA Board (DE1-SoC, DE10-Lite), ESP32, Arduino, Function Generator

PROJECTS

FPGA Tron- Light-Cycle Game | *C, DE1-SoC, Nios V, Makefile*

- Implemented VGA arena running on a Nios V (RISC-V) soft processor on the DE1-SoC FPGA in bare-metal C
- Interrupt-driven player input, using pushbutton queues, tick and speed controlled via switches
- Lookahead AI opponent, collision with any obstacle ends a round and simultaneous collision is a draw
- Score displayed live on 7-segment HEX display; screen colors red when game ends

FPGA Digital Clock | *SystemVerilog, Quartus, Questa, ModelSim, DE1-SoC*

- Developed a 24-hour format clock on the DE1-SoC board using SystemVerilog
- Implemented a feature to allow the user to change the hours and minutes respectively using a Moore FSM
- User input was handled using a two-stage synchronizer and a positive edge detector
- Used testbenching for the modules implemented for hardware verification and validation

Discrete Transistor 7-Segment Decoder | *LTSpice, Quartus, ModelSim, Falstad*

- Utilized truth tables, Karnaugh maps and boolean simplifications to create digital circuits for each output
- Implemented NOR gates only using 40+ discrete 2N222 NPN transistors to optimize cost, space and simplicity
- Verified the design rigorously through Quartus and LTSpice using RTL logic, integrating all digital circuits
- Calculated base resistors to ensure transistors operated in the saturation region for consistent LED brightness

RISC-V Single-Cycle CPU | *SystemVerilog, RISC-V Assembly, ModelSim, DE1-SoC*

- Single-cycle architecture: every instruction complete in one clock cycle, no pipelining or caching
- Controller split into a main and ALU decoder, directing a datapath with a 32×32-bit register file, ALU, and mux
- Supports R-type (add, sub, and, or, slt), I-type (lw, addi, andi, ori, slti), S-type, B-type, and J-type instructions
- DE-board peripherals (LEDs, HEX displays, slide switches) exposed via MMIO at 0xFF200000

WORK EXPERIENCE

EPS Software Team Member

UBC Orbit

Jan. 2026 – Present

Vancouver, BC

- Contribute to refining and maintaining our Python-based dynamic power model for SIL testing
- Establish a precise energy budget through iterative simulations, modeling solar generation and battery logic
- Collaborate with other subsystems to validate data and optimize for a positive power balance during orbit

Vice-chair & Events Team Member

UBC IEEE Quantum Technical Society

Sep. 2025 – Present

Vancouver, BC

- Participated in industry networking events, building connections with professionals in various companies
- Designed and delivered hands-on workshops on Quantum Computing, improving participants' technical proficiency
- Organized several technical events throughout the semester, while promoting IEEE and its core values